

Adjoint Map and Killing form

Defⁿ [Adjoint]

$(L, [-, -])$ a complex Lie algebra and $h \in L$, then define adjoint map w.r.t. $h \in L$.

$$\left. \begin{aligned} \text{ad}(h) : L &\xrightarrow{\sim} L \quad (\text{linear map}). \\ l &\mapsto \text{ad}(h)(l) := [h, l] \end{aligned} \right\} \Rightarrow \boxed{\text{ad}(h) := [h, -]}$$

Defⁿ [Killing Form]

The bilinear form $K: L \times L \xrightarrow{\sim} \mathbb{C}$ $L \xrightarrow{\sim} L$ (thus trace is well-defined).

$$\boxed{K(a, b) = \text{tr}(\text{ad}(a) \circ \text{ad}(b))} \quad \forall a, b \in L.$$

is called the Killing form

Remark:

(1) L is finite-dim vector-space $\Rightarrow \text{tr}$ is cyclic

$$\Rightarrow K(a, b) = K(b, a) \quad [\text{Symmetric}].$$

(2) L semi-simple iff K is non-degenerate:

$$\forall a \in L : K(a, b) = 0 \Rightarrow b = 0.$$

$\Rightarrow K$ is a pseudo-inner product on semi-simple L .

⊗ Now consider, for actual calculation, components of $\text{ad}(h)$ and K w.r.t. a basis

$$E_1, \dots, E_{\dim L} \quad \text{of } L$$



$$\{\text{dual basis}\} \quad \varepsilon^1, \dots, \varepsilon^{\dim L} \quad \text{of } L^*$$

$$\text{Then } \underbrace{\text{ad}(E_i)^j}_K := \varepsilon^j(\text{ad}(E_i)(E_k)) = \varepsilon^j([E_i, E_k])$$

$$\text{End}(L) \cong_{\text{vec}} T_1^* L$$

$$\text{since } E_i, E_k \in L \Rightarrow [E_i, E_k] \in L$$

structure const.
of L w.r.t.
chosen basis.

$$\rightarrow [E_i, E_k] = C^m_{ik} E_m$$

expanded in
the same basis

$$\Rightarrow \text{ad}(E_i)^j_k = \varepsilon^j(C^m_{ik} E_m) = C^m_{ik} \varepsilon^j(E_m) = C^m_{ik} \delta^j_m = C^j_{ik}$$

Killing form:

$$\underbrace{K(E_i, E_j)}_{K_{ij}} = \text{tr}(\text{ad}(E_i) \circ \text{ad}(E_j)) = [\text{ad}(E_i) \circ \text{ad}(E_j)]^m_m = \text{ad}(E_i)^m_n \text{ad}(E_j)^n_m = C^m_{in} \cdot C^n_{jm}$$

$$\Rightarrow \boxed{K_{ij} := C^m_{in} C^n_{jm}}$$

⊛ A linear map $\varphi: V \rightarrow V$ is called (anti-) symmetric w.r.t. a non-degenerate bilinear form B .
 if $B(\varphi(v), w) = \pm B(v, \varphi(w)) \quad \forall v, w \in V.$ + $\rightarrow$ symmetric
 - $\rightarrow$ anti-symmetric.

⊛ Let $(L, [\cdot, \cdot])$ be a semi-simple Lie algebra $\Rightarrow K$ is a pseudo-inner product.
 and for any $h \in L$, $\text{ad}(h)$ is anti-symmetric

Now we are ready to actually answer the question of classification.

Defⁿ [Cartan Subalgebra]

$(L, [-, -])$ is a Lie algebra. Then a Cartan subalgebra H is

⊙ $H \subseteq L$ is subspace.

⊙ H is maximal subalgebra of L such that there exists a basis of H

$$h_1, \dots, h_m, e_1, \dots, e_{d-m} \quad ; \quad d = \dim L.$$

that can be extended to a basis of L , such that $e_1, \dots, e_{d-m}$ are eigenvector for any $\text{ad}(h)$ where $h \in H$.

$$\text{ad}(h) e_\alpha = \underbrace{\lambda_\alpha(h)}_{\in \mathbb{C}} e_\alpha \quad ; \quad \alpha = 1, \dots, d-m$$

(14.1)

Theorem: Any finite-dim Lie algebra posses a Cartan subalgebra.

Theorem: (14.2) L is simple $\Rightarrow H$ abelian i.e., $[H, H] = 0$.

Observation: $\text{ad}(h) e_\alpha = [h, e_\alpha] = \lambda_\alpha(h) e_\alpha \} \Rightarrow$ linear in $h \in H$

$$\Rightarrow \lambda_\alpha: H \rightarrow \mathbb{C}$$

$$\Leftrightarrow \lambda_\alpha \in H^*$$

Defⁿ [Roots]

The $\lambda_1, \dots, \lambda_{d-m} \in H^*$ are called the roots of the Lie algebra. The set of all roots are

$$\Phi := \{\lambda_1, \dots, \lambda_{d-m}\} \subseteq H^*$$

is called the root set.

Remarks: (1) $\text{ad}(h)$ is anti-sym. w.r.t. Killing form $\Rightarrow \dots$

$$\dots \Rightarrow \lambda \in \Phi \Rightarrow -\lambda \in \Phi.$$

(2) Φ is not linearly independent.

Defⁿ [Fundamental Roots]

The set of fundamental roots $\Pi \subset \Phi$ such that

(a) Π is linearly independent ; $\Pi = \{\pi_1, \dots, \pi_f\}$

(b) $\forall \lambda \in \Phi : \exists n_1, \dots, n_f \in \underbrace{\mathbb{N}_0}_{\text{non-negative integer}} \exists \varepsilon \in \{+1, -1\} : \lambda = \varepsilon \sum_{i=1}^f n_i \pi_i$

Fact: such $\Pi \subset \Phi$ can always be found.

Largest proper subalg.
 (w.r.t. $\subseteq$).

Maximal Subalgebra

Let M is maximal

then for any subalgebra

$$S: M \subseteq S \subseteq L$$

$$\Downarrow$$

$$M=S \text{ or } S=L.$$

$\dim H$ is known
 as rank of L .

Theorem (14.3):

$$\text{span}_{\mathbb{C}}(\Pi) = H^*$$

⊗ Remark: Π is not unique.

Defⁿ

(1) $H_{\mathbb{R}}^* = \text{span}_{\mathbb{R}}(\Pi)$

Then we have

$$\Pi \subset \underbrace{\Phi}_{\text{span}_{\pm\mathbb{N}}(\Pi)} \subset H_{\mathbb{R}}^* \subset H^*$$

Φ only contains certain signed linear combination of Π .

(2) have $K: L \times L \xrightarrow{\sim} \mathbb{C}$
 $\Rightarrow K|_H: H \times H \xrightarrow{\sim} \mathbb{C}$ } want a Killing form K^* on H^* .

Recall for finite-dim

$$\begin{aligned} i: H &\xrightarrow{\sim} H^* \\ i(h) &:= K(h, \cdot) \in \mathbb{C} \end{aligned} \quad \left. \vphantom{\begin{aligned} i: H &\xrightarrow{\sim} H^* \\ i(h) &:= K(h, \cdot) \in \mathbb{C} \end{aligned}} \right\} \begin{array}{l} \text{Pseudo-inner product} \\ \text{induces this isomorphism.} \end{array}$$

since K is non-deg. $\Rightarrow i^{-1}$ exists

$\Downarrow$
 L is semi-simple.

With is $K^*: H^* \times H^* \rightarrow \mathbb{C}$

$$K^*(\mu, \nu) := K(i^{-1}(\mu), i^{-1}(\nu)).$$

Theorem (14.4):

$$K^*: H_{\mathbb{R}}^* \times H_{\mathbb{R}}^* \rightarrow \mathbb{R} \quad \text{and} \quad K^* \text{ is positive definite on } H_{\mathbb{R}}^*.$$

$$K^*(\alpha, \alpha) \geq 0 \quad \forall \alpha \in H_{\mathbb{R}}^* \quad (\text{equality holds only for } \alpha = 0).$$

$\Rightarrow$ On $H_{\mathbb{R}}^*$ we can calculate lengths & angle.

In particular, one can calculate length of & angle b/w fundamental roots.

(since $\Phi \subset H_{\mathbb{R}}^*$ we can do this for any root).

⊗ Now, we are almost there to actually answer the classification problem. In this process, we have reduced the whole problem to Π . As

$$\Pi \longrightarrow H^* = \text{span}_{\mathbb{C}}(\Pi) \xrightarrow{\cong_{\text{vec}}} H \xrightarrow[\text{subalgebra}]{\text{cartan}} L$$

However, it is not clear that can we recover Φ from Π .

Defⁿ [Weyl transformation].

For any $\lambda \in \Phi$ define

$$S_{\lambda}: H_{\mathbb{R}}^* \xrightarrow{\sim} H_{\mathbb{R}}^* \quad ; \quad S_{\lambda}(\mu) := \mu - 2 \frac{K^*(\lambda, \mu)}{K^*(\lambda, \lambda)} \lambda$$

S_{λ} is known as Weyl transf.

! linear in μ ,
 non-linear in λ .

and $W := \{s_\lambda : \lambda \in \Phi\}$ called Weyl group. with group op^r being composition of map.

Theorem

(1) "The Weyl group is generated by the fundamental root in Π ":

$$\forall \omega \in W : \exists \pi_1, \dots, \pi_n \in \Pi : \omega = s_{\pi_1} \circ s_{\pi_2} \circ \dots \circ s_{\pi_n}$$

(2) Every root $\lambda \in \Phi$ can be produced from a fundament root $\pi \in \Pi$. by action of the Weyl grp.

$$\forall \lambda \in \Phi \exists \omega \in W, \pi \in \Pi : \lambda = \omega(\pi)$$

(3) The Weyl group permutes the roots:

$$\forall \omega \in W, \forall \lambda \in \Phi : \omega(\lambda) \in \Phi.$$

Consider $\pi_i, \pi_j \in \Pi$.

$$\Phi \ni s_{\pi_i}(\pi_j) = \pi_j - 2 \frac{k^*(\pi_i, \pi_j)}{k^*(\pi_i, \pi_i)} \pi_i \quad \text{for } i \neq j \Rightarrow -2 \frac{k^*(\pi_i, \pi_j)}{k^*(\pi_i, \pi_i)} \in \mathbb{N}_0.$$

(as $\pi_j \in \Phi$)

Define [Cartan Matrix]

$$C_{ij} := 2 \frac{k^*(\pi_i, \pi_j)}{k^*(\pi_i, \pi_i)} \quad \begin{matrix} \nearrow C_{ii} = 2 \\ \searrow C_{ij} \in \mathbb{Z}_{\leq 0} \text{ (non-positive integers)} \end{matrix}$$

⊗ $C_{ij} \neq C_{ji}$ iff $C_{ij} \neq 0$. $\Rightarrow$ Cartan matrix is not symmetric in general.

Defⁿ [Bond Number]

$$n_{ij} = C_{ij} C_{ji} \quad (\text{no summation}).$$

$$= 4 \frac{k^*(\pi_i, \pi_j)}{k^*(\pi_i, \pi_i)} \cdot \frac{k^*(\pi_j, \pi_i)}{k^*(\pi_j, \pi_j)} = 4 \left(\frac{k^*(\pi_i, \pi_j)}{|\pi_i| |\pi_j|} \right)^2 = 4 \cos^2 \varphi_{ij}$$

⊗ φ_{ij} is angle b/w fundamental roots π_i, π_j .

Observe : $C_{ij} \in \mathbb{Z}_{\leq 0} \Rightarrow n_{ij} \in \mathbb{N}_0$

and $0 \leq n_{ij} \leq 4$ (for $n_{ij} = 4 \Rightarrow C_{ij} = 2 \Rightarrow i=j$)

Thus equality with 4 is trivial case. $\Rightarrow n_{ij} \in \{0, 1, 2, 3\}$ for $i \neq j$

C_{ij}	C_{ji}	n_{ij}
0	0	0
-1	-1	1
$\begin{Bmatrix} -1 \\ -2 \end{Bmatrix}$	$\begin{Bmatrix} -2 \\ -1 \end{Bmatrix}$	2
$\begin{Bmatrix} -1 \\ -3 \end{Bmatrix}$	$\begin{Bmatrix} -3 \\ -1 \end{Bmatrix}$	3

$\nearrow$ we can show that $|\pi_i| < |\pi_j|$ or $|\pi_i| > |\pi_j|$

Defⁿ [Dynkin Diagrams].

- (i) for every fundamental root draw a circle. $\circ$
- (ii) if $\circ$ $\circ$ represents $\pi_i, \pi_j \in \Pi$, draw n_{ij} lines b/w them
- (iii) if there are 2 or 3 lines b/w two roots, use a $<$ sign to relate the roots:

$$\begin{array}{ccc} \circ & \xrightarrow{\quad} & \circ \\ \pi_i & & \pi_j \\ |\pi_i| & > & |\pi_j| \end{array}$$

Theorem [Cartan, Killing]

Any finite-dim simple $\mathbb{C}$ -Lie algebra can be reconstructed from its set of fundamental roots Π and the latter only come in the following form.

(i) Infinite Families:

$$\odot A : \begin{array}{c} \circ, \quad \circ - \circ, \quad \circ - \circ - \circ, \quad \dots, \quad \circ - \circ - \dots - \circ \\ A_1 \quad A_2 \quad A_3 \quad A_n \end{array} \quad \left\} A_n ; n \geq 1\right.$$

$$\odot B : \begin{array}{c} \circ \rightleftharpoons \circ, \quad \circ - \circ \rightleftharpoons \circ, \quad \dots, \quad \circ - \dots - \circ \rightleftharpoons \circ \\ B_2 \quad B_3 \quad B_n \end{array} \quad \left\} B_n ; n \geq 2\right.$$

$$\odot C : \begin{array}{c} \circ - \circ \leftrightsquigarrow \circ, \quad \dots, \quad \circ - \dots - \circ \leftrightsquigarrow \circ \\ C_3 \quad C_n \end{array} \quad \left\} C_n ; n \geq 3\right.$$

$$\odot D : \begin{array}{c} \circ - \circ \begin{array}{l} \nearrow \circ \\ \searrow \circ \end{array}, \quad \circ - \dots - \circ \begin{array}{l} \nearrow \circ \\ \searrow \circ \end{array} \\ D_4 \quad D_n \end{array} \quad \left\} D_n ; n \geq 4\right.$$

(ii) Exceptional cases: $\frac{5}{}$

$$\odot E_6 : \begin{array}{c} \circ \\ | \\ \circ - \circ - \circ - \circ - \circ \end{array}$$

$$\odot E_7 : \begin{array}{c} \circ - \circ - \circ - \circ - \circ \\ | \\ \circ \end{array}$$

$$\odot E_8 : \begin{array}{c} \circ - \circ - \circ - \circ - \circ - \circ - \circ \\ | \\ \circ \end{array}$$

$$\odot F_4 : \circ - \circ \leftrightsquigarrow \circ - \circ$$

$$\odot G_2 : \circ \leftrightsquigarrow \circ$$